London South Bank University

ENGINEERING PORTFOLIO / 2025 – 26

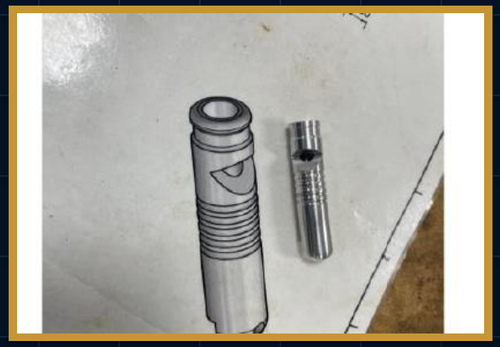
WHISTLE
MANUFACTURING

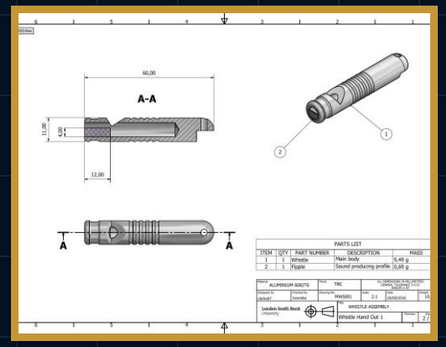
From raw aluminium to a precision-machined whistle

FIRST-YEAR MECHANICAL ENGINEERING WORKSHOP PROJECT

Turning / Boring / Milling / Inspection









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INDIVIDUAL WORKSHOP PROJECT

Conventional machining and practical skills

SEEKING A 2027

INDUSTRIAL PLACEMENT

Project at a glance



WORKSHOP PART AND ENGINEERING DRAWING

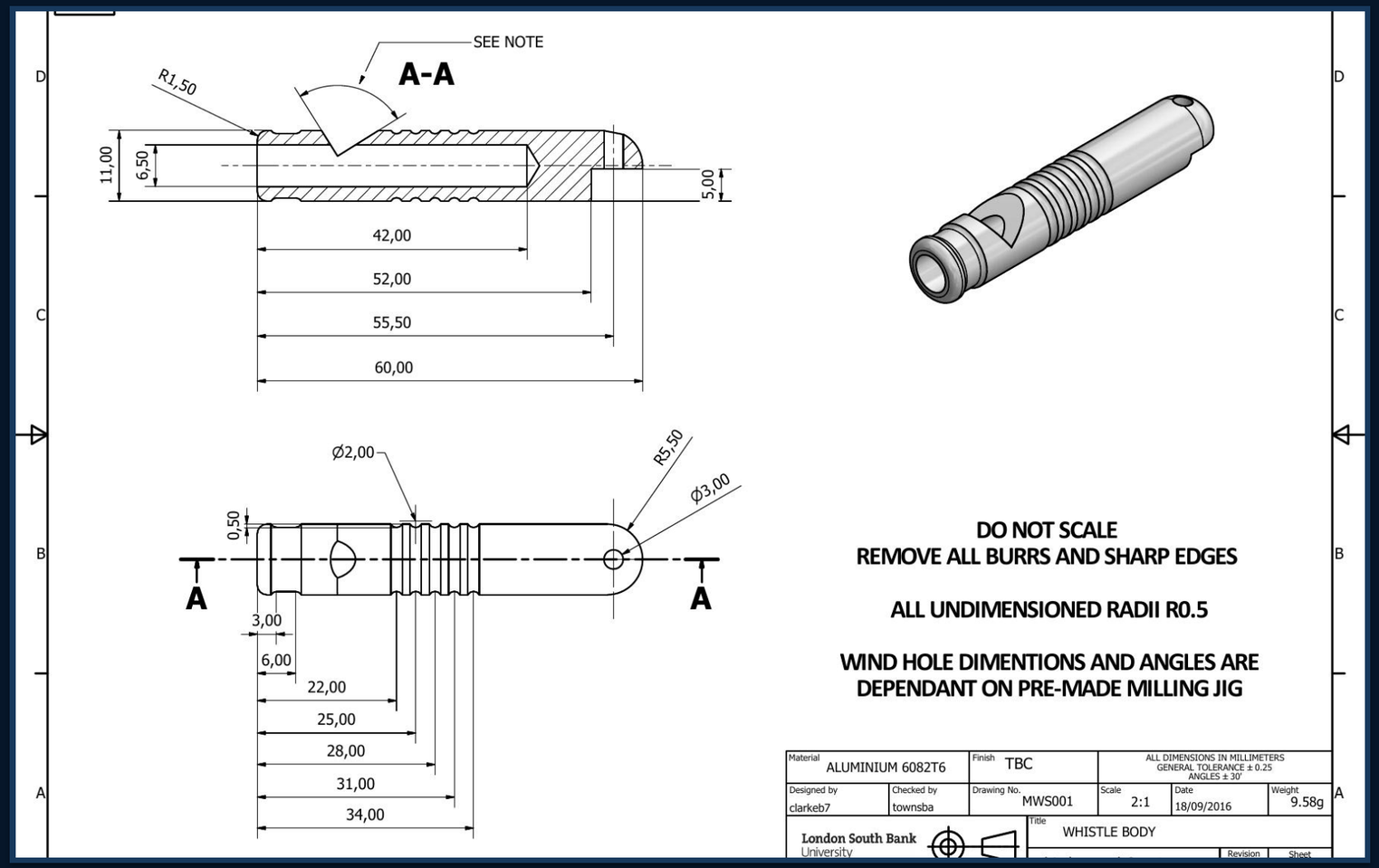
THE BRIEF

Manufacture a metal whistle from aluminium stock using conventional workshop equipment, following a supplied engineering drawing.

Core operations

- Facing and turning
- Drilling / boring
- Groove machining
- Vertical milling
- Filing, fitting and inspection

Reading the engineering drawing



SOURCE: LSBU WHISTLE HANDOUT, BODY DRAWING

DESIGN REQUIREMENTS

60 mm overall length

11 mm external diameter

Aluminium 6082-T6 specified

General tolerance ±0.25 mm

Remove burrs and sharp edges

The wind-hole geometry depends on the pre-made milling jig.

Manufacturing plan: stock to component

01

Prepare stock

02

Face & turn

03

Bore the body

04

Machine grooves

05

Mill features

06

File & inspect

Starting material & workshop safety



APPROX. 90 mm ALUMINIUM STOCK, FROM WORKSHOP RECORD



LATHE SETUP DURING PRACTICAL SESSION

I worked under workshop supervision and followed the instructed machine-safety procedures.

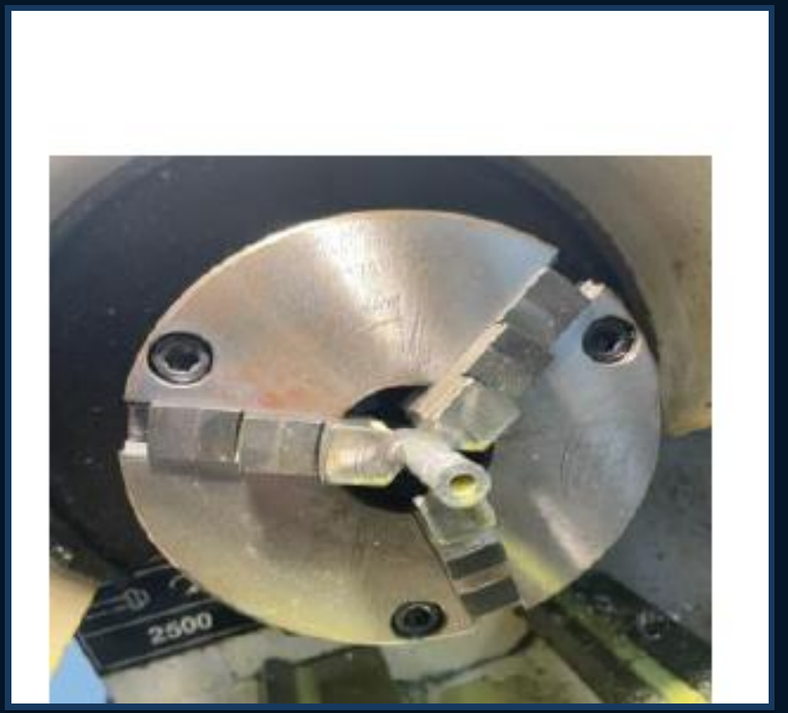
Lathe work: facing, turning and boring



LATHE OPERATION



AFTER TURNING



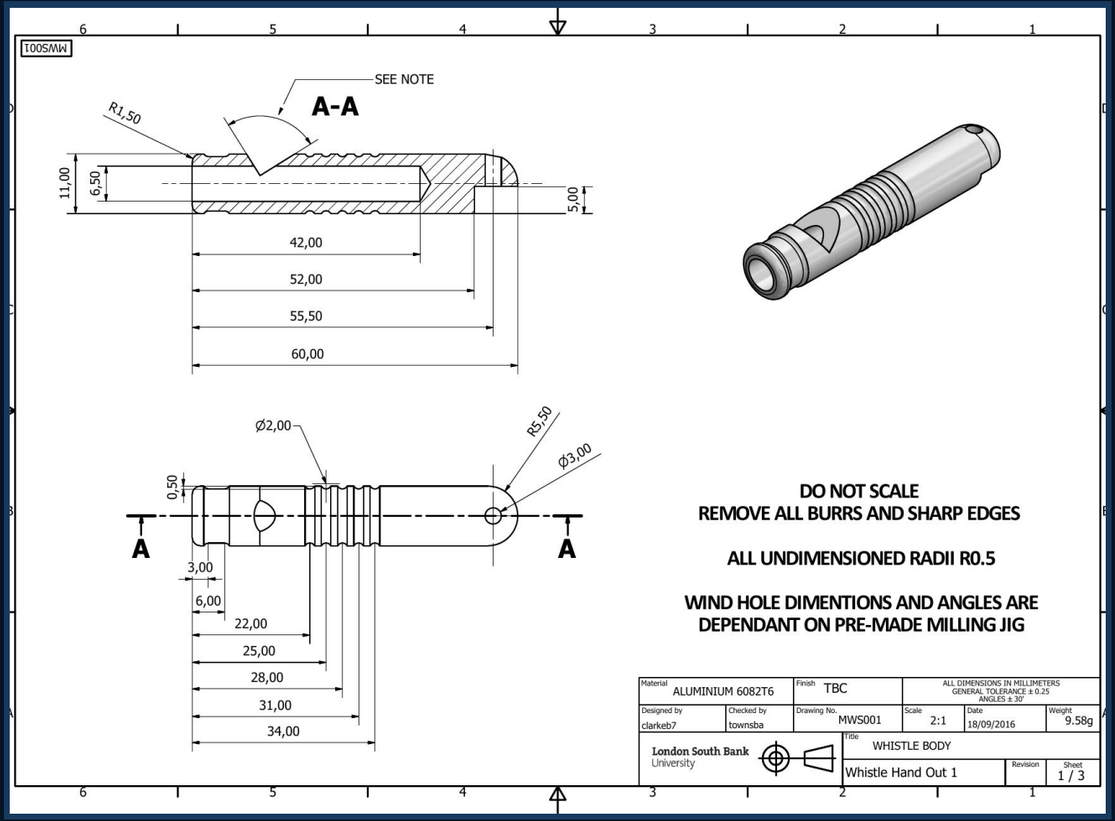
BORING STAGE

The workshop poster records turning to approximately 11 mm outside diameter and boring to approximately 7 mm. Early inaccuracies required adjustment and rechecking.

Machining the grooves



GROOVE / ROUNDING OPERATION



GROOVE POSITIONS FROM TECHNICAL DRAWING

Workshop notes describe four groove locations around 3–6, 22–23, 25–26 and 28–29 mm. Positions were checked against the drawing.

Vertical milling: functional geometry



VERTICAL MILLING MACHINE



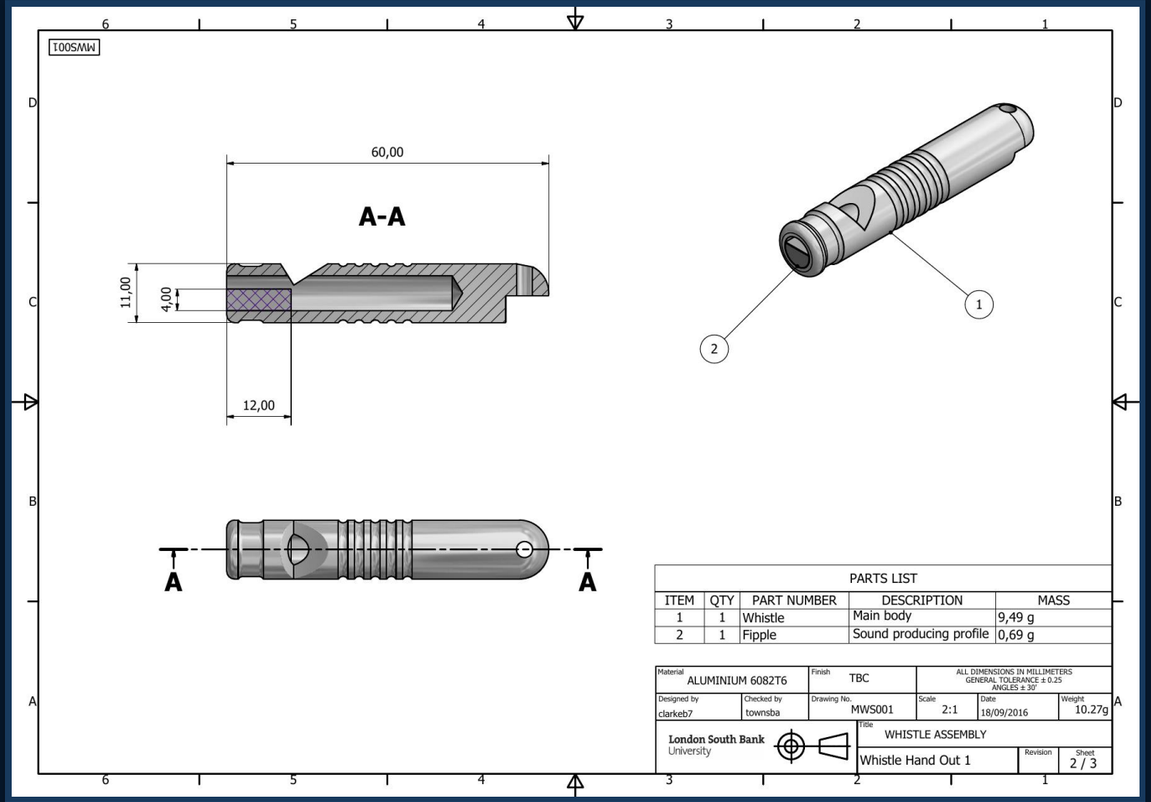
CUTTING THE WHISTLE FEATURE

The documented sequence includes milling a flat feature and creating the V-shaped wind opening. Careful setup and alignment were essential.

Fipple fitting & manual finishing



MANUAL FILING / FITTING



TWO-PART WHISTLE ASSEMBLY

The fipple (sound-producing insert) was manually fitted. The drawing identifies the whistle body and fipple as separate parts.

Inspection & dimensional checks



FINAL LENGTH CHECK



FINISHED COMPONENT COMPARED WITH DRAWING

The poster records a final length of 60 mm and width / diameter of 11 mm. These are reported workshop measurements, not a certified tolerance assessment.

Challenges, corrections and learning

CHALLENGES

- Interpreting detailed mechanical drawings
- Achieving precision during turning
- Manual filing and fitting
- Recovering from early machining mistakes

HOW I RESPONDED

- Rechecked dimensions against the handout
- Adjusted machining and finishing steps
- Compared the part with the drawing
- Asked for instructor feedback and inspection

What this project demonstrates

Manufacturing	Lathe and vertical milling experience
Technical literacy	Reading orthographic and sectional views
Metrology	Dimensional measurement and inspection
Problem-solving	Iterative corrections during manufacture
Professional practice	Workshop safety and process planning

Reflection & next steps



COMPLETED WORKSHOP COMPONENT

WHAT I WOULD IMPROVE

- Plan machining stages more systematically
- Record intermediate dimensions in a process sheet
- Improve repeatability and surface finish
- Strengthen measurement and metrology skills

A practical introduction to translating design intent into a manufactured part.

Thank you | Let’s connect



BOKHTEAR UDDIN BADHON

BEng (Hons) Mechanical Engineering

London South Bank University

Seeking a 2027 industrial placement in mechanical engineering, manufacturing, design or related engineering roles.

Portfolio project: Whistle Manufacturing • Technical drawings • Workshop practice